

The Analysis and Design of Linear Circuits

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 Chapter Basic Circuit Analysis
 Problem 2.68

Exercise 2-64

Ideally, a voltmeter has infinite internal resistance and can be placed across any device to read the voltage without affecting the result. A particular digital multimeter (DMM), a common laboratory tool, is connected across the circuit shown in figure 1. The expected voltage was 10.2 V. However, the DMM reads 7.61 V. The large, but finite, internal resistance of the DMM was "loading" the circuit and causing a wrong measurement to be made. Find the value of the internal resistance R_M of this DMM.

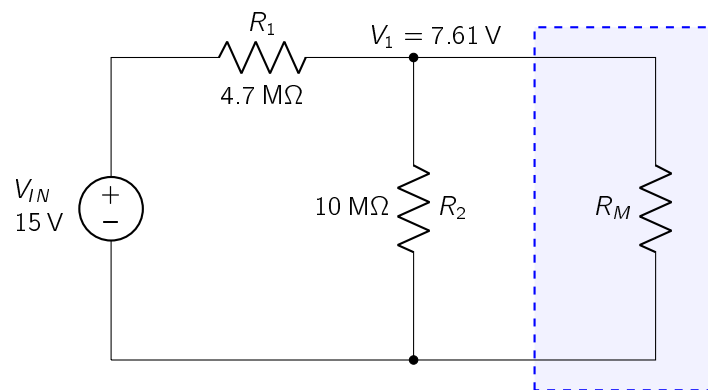


Figure 1:

Answer

The unknown resistor R_M is in parallel with R_2 resistor. Since voltages across parallel elements are equal, the voltage V_1 appears across both. We first define an equivalent resistance $R_{EQ} = R_M || R_2$ as

$$R_{EQ} = \frac{R_2 R_M}{R_2 + R_M} \quad (1)$$

We write the voltage division rule in terms of R_{EQ} as

$$V_1 = \frac{R_{EQ}}{R_{EQ} + R_1} V_{IN} = 7.61 \text{ V} \quad (2)$$

Solving equation (2) for R_{EQ} yields

$$\begin{aligned} R_{EQ} V_{IN} &= V_1 (R_{EQ} + R_1) \\ R_{EQ} V_{IN} &= R_{EQ} V_1 + R_1 V_1 \\ (V_{IN} - V_1) R_{EQ} &= R_1 V_1 \\ R_{EQ} &= \frac{R_1 V_1}{V_{IN} - V_1} \end{aligned} \quad (3)$$

Therefore

$$R_{EQ} = 4.84 \text{ M}\Omega \quad (4)$$

Then, we substitute the value presented in (4) on equation (1) and solving for R_2

$$\begin{aligned}R_2 R_M &= R_{EQ}(R_2 + R_M) \\R_2 R_M &= R_2 R_{EQ} + R_M R_{EQ} \\(R_2 - R_{EQ})R_M &= R_2 R_{EQ}\end{aligned}$$

$$R_M = \frac{R_2 R_{EQ}}{R_2 - R_{EQ}} \quad (5)$$

so

$$R_M = \frac{(10 \text{ M}\Omega)(4.84 \text{ M}\Omega)}{10 \text{ M}\Omega - 4.84 \text{ M}\Omega} = 9.3798 \text{ M}\Omega \quad (6)$$